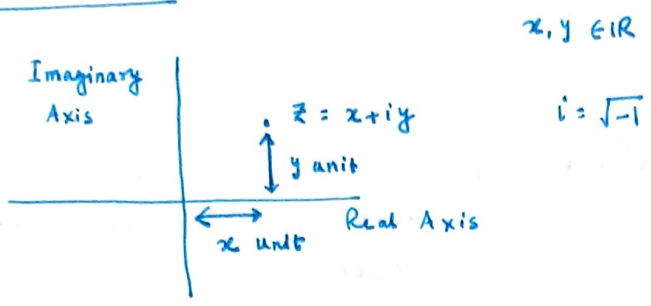
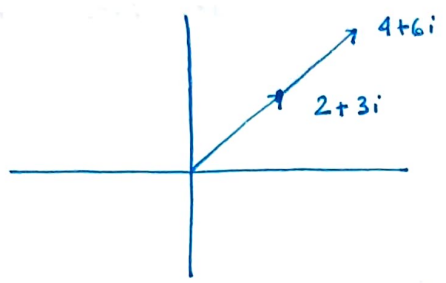


Complex Numbers



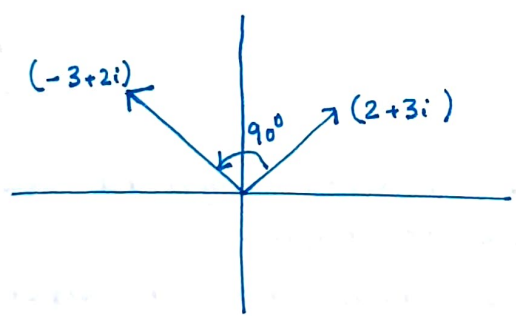
Multiplication by a real number : $(2 + 3i) \xrightarrow{\times 2} (4 + 6i)$



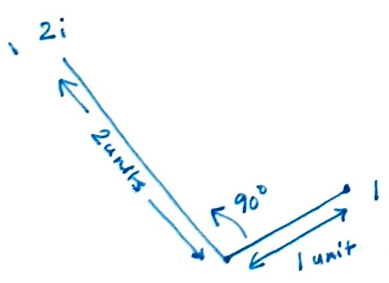
Scale up/down.
Same direction about the origin

Multiplication by a purely imaginary number:

$$(2 + 3i) \xrightarrow{\times i} (-3 + 2i)$$



Rotation by 90°
about the origin

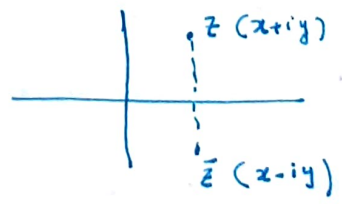


$$1 \xrightarrow{\times 2i} 2i$$

Rotation + Scale.

Conjugate :

$\bar{z} = x - iy$
Reflection about
Real axis.



$$x = \text{Re}(z) = \text{Real Part of } z = \frac{z + \bar{z}}{2}$$

$$y = \text{Im}(z) = \frac{z - \bar{z}}{2i}$$

- z is purely real $\Leftrightarrow \text{Im}(z) = 0 \Leftrightarrow \frac{z - \bar{z}}{2i} = 0 \Leftrightarrow \bar{z} = z$
- z is purely imaginary $\Leftrightarrow \bar{z} = -z$.

Connection with Matrices:

- Symmetric Matrix : $A^T = A \rightarrow$ all the eigenvalues of a real symmetric matrix are purely real.
- Skew-symmetric Matrix : $A^T = -A \rightarrow$ All the eigenvalues of a real skew-symmetric matrix are purely imaginary or 0.

Eigenvalues:

$A_{n \times n}$

eigenvalues are the solutions of the polynomial equation

$$\det(A - \lambda I) = 0$$

Why? $\rightarrow$ let $\underline{v}$ be the corresponding eigenvector with eigenvalue λ .

So, $A\underline{v} = \lambda\underline{v} \quad (\underline{v} \neq \underline{0})$

$\Rightarrow (A - \lambda I)\underline{v} = \underline{0}$; Since $\underline{v} \neq \underline{0}$, $A - \lambda I$ must be singular

Because if $A - \lambda I$ would exist, then $\underline{v} = (A - \lambda I)^{-1} \underline{0} = \underline{0}$, contradiction.

So, $A - \lambda I$ is singular $\Rightarrow \det(A - \lambda I) = 0$.

The eqn $\det(A - \lambda I) = 0$ is a n -degree polynomial equation, so it ~~must~~ has n roots, real or complex.

Hence every $n \times n$ matrix has n eigenvalues, real or complex.

One example of calculating eigenvalues:

Let, $A = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$. Find the eigenvalues of A .

Ans: To get the eigenvalues, we have to solve the equation:

$$\det(A - \lambda I) = 0$$

$$\Rightarrow \det\left(\begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = 0$$

$$\Rightarrow \det \left(\begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right) = 0$$

$$\Rightarrow \det \left(\begin{bmatrix} 2-\lambda & 3 \\ 3 & 4-\lambda \end{bmatrix} \right) = 0$$

$$\Rightarrow (2-\lambda)(4-\lambda) - 9 = 0$$

$$\Rightarrow \lambda^2 - 6\lambda + 8 - 9 = 0$$

$$\Rightarrow \lambda^2 - 6\lambda - 1 = 0 \quad \Rightarrow \lambda = \frac{6 \pm \sqrt{36+4}}{2}$$

$$= 3 + \sqrt{10}, \quad 3 - \sqrt{10}$$

So, the two eigenvalues are: $(3 + \sqrt{10}), (3 - \sqrt{10})$

□ Observe that: i) The sum of the eigenvalues

$$= (3 + \sqrt{10}) + (3 - \sqrt{10})$$

$$= 6$$

$$= \text{Trace}(A) = \text{Sum of diagonal entries of } A = 3 + 3$$

ii) Product of the eigenvalues

$$= (3 + \sqrt{10})(3 - \sqrt{10})$$

$$= 9 - 10$$

$$= -1 = \det(A) = 8 - 9 = -1$$

□ In any square matrix, sum of eigenvalues = Trace of the matrix.

and, product of eigenvalues = determinant of the matrix.

□ The previous one was an example of a symmetric matrix having real eigenvalues. Now we illustrate an example with a skew-symmetric matrix.

$$\square \text{ Let, } A = \begin{bmatrix} 0 & -3 \\ 3 & 0 \end{bmatrix} ; \text{ Note that } A^T = \begin{bmatrix} 0 & 3 \\ -3 & 0 \end{bmatrix} \\ = - \begin{bmatrix} 0 & -3 \\ 3 & 0 \end{bmatrix} = -A.$$

A is skew-symmetric. Find the eigenvalues of A .

Ans: We have to solve

$$\begin{aligned}\det(A - \lambda I) &= 0 \\ \Rightarrow \det \left(\begin{bmatrix} 0 & -3 \\ 3 & 0 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right) &= 0 \\ \Rightarrow \det \left(\begin{bmatrix} -\lambda & -3 \\ 3 & -\lambda \end{bmatrix} \right) &= 0 \\ \Rightarrow \lambda^2 + 9 &= 0 \\ \Rightarrow \lambda^2 &= -9 \quad \Rightarrow \lambda = \pm 3i\end{aligned}$$

So, the eigenvalues are $3i, -3i$ (purely imaginary)

□ Any odd-ordered skew symmetric matrix has at least one eigenvalue $= 0$.

Pf: Let $A_{n \times n}$ be skew symmetric. (n is odd)

$$\text{So, } A^T = -A$$

$$\Rightarrow \det(A^T) = \det(-A)$$

$$\Rightarrow \det(A) = (-1)^n \det(A) \quad \left[\because \det(A^T) = \det(A), \right. \\ \left. \det(cA) = c^n \det(A) \right]$$

$$\Rightarrow \det(A) = -\det(A) \quad [\text{As } n \text{ is odd}]$$

$$\Rightarrow 2 \det(A) = 0$$

$$\Rightarrow \det(A) = 0$$

$$\Rightarrow \text{product of eigenvalues of } A = 0$$

$$\Rightarrow \text{At least one of the eigenvalues of } A \text{ is } 0. \quad \square$$

Quadratic Form

The quadratic form of a symmetric matrix $A_{n \times n}$ is defined as

$$Q(\underline{x}) = \underline{x}^T A \underline{x} \quad (\text{where } \underline{x} \text{ is any } n \times 1 \text{ vector})$$

Example: Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} a & h \\ h & b \end{bmatrix}$

($a_{12} = a_{21}$, as A is symmetric)

So, for $\underline{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$, the quadratic form:

$$Q(\underline{x}) = \underline{x}^T A \underline{x}$$

$$\begin{aligned}
 \Rightarrow Q(x_1, x_2) &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a & h \\ h & b \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\
 &= \begin{bmatrix} ax_1 + hx_2 & hx_1 + bx_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\
 &= ax_1^2 + hx_1x_2 + hx_1x_2 + bx_2^2 \\
 &= ax_1^2 + 2hx_1x_2 + bx_2^2
 \end{aligned}$$

[So], For $A = \begin{bmatrix} a & h \\ h & b \end{bmatrix}$, the quadratic form = $ax_1^2 + 2hx_1x_2 + bx_2^2$

conversely, for the quadratic expression

$$Q(x_1, x_2) = ax_1^2 + 2hx_1x_2 + bx_2^2, \text{ the}$$

Corresponding symmetric matrix is $\rightarrow$

$$A = \begin{bmatrix} a & h \\ h & b \end{bmatrix}$$

[ex] i) Let $Q(x_1, x_2) = 3x_1^2 + 5x_1x_2 + 4x_2^2$, the
corresponding symmetric matrix is $A = \begin{bmatrix} 3 & 5/2 \\ 5/2 & 4 \end{bmatrix}$

ii) Let $Q(x_1, x_2, x_3) = 3x_1^2 + 4x_2^2 + 2x_3^2 + 6x_1x_2 + 8x_2x_3 + 4x_3x_1$

the corresponding symmetric matrix is

$$A = \begin{bmatrix} 3 & 3 & 2 \\ 3 & 4 & 4 \\ 2 & 4 & 2 \end{bmatrix}$$

for symmetric matrix A,

[Results]

i) $\text{Max}_{\|x\|=1} x^T A x = \text{Largest eigenvalue of } A$.

ii) $\text{Min}_{\|x\|=1} x^T A x = \text{Smallest eigenvalue of } A$.

$$[\text{Let, } x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \|x\| = \sqrt{x^T x} = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}]$$

Application: Let $x_1, x_2 \in \mathbb{R}$, satisfying $x_1^2 + x_2^2 = 1$.

Find the maximum and the minimum value of $3x_1^2 + 4x_1x_2 + 5x_2^2$.

Ans: Let $\underline{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $\underline{x}^T \underline{x} = \|\underline{x}\|^2 = x_1^2 + x_2^2 = 1$, (given)
 $\Rightarrow \|\underline{x}\| = 1$.

Now, $Q(\underline{x}) = 3x_1^2 + 4x_1x_2 + 5x_2^2$ is a quadratic form with the corresponding matrix (symmetric) $\rightarrow$

$$A = \begin{bmatrix} 3 & 2 \\ 2 & 5 \end{bmatrix}$$

i.e. $Q(\underline{x}) = \underline{x}^T A \underline{x} = 3x_1^2 + 4x_1x_2 + 5x_2^2$

Hence, by previous result,

$$\max_{\|\underline{x}\|=1} \underline{x}^T A \underline{x} = \text{largest eigenvalue of } A$$

$$\min_{\|\underline{x}\|=1} \underline{x}^T A \underline{x} = \text{smallest eigenvalue of } A.$$

To find the eigenvalues of A : $\det(A - \lambda I) = 0$

$$\Rightarrow \det \left(\begin{bmatrix} 3 & 2 \\ 2 & 5 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right) = 0$$

$$\Rightarrow \det \left(\begin{bmatrix} 3-\lambda & 2 \\ 2 & 5-\lambda \end{bmatrix} \right) = 0$$

$$\Rightarrow (3-\lambda)(5-\lambda) - 4 = 0$$

$$\Rightarrow \lambda^2 - 8\lambda + 15 - 4 = 0$$

$$\Rightarrow \lambda^2 - 8\lambda + 11 = 0$$

$$\Rightarrow \lambda = \frac{8 \pm \sqrt{64 - 44}}{2}$$

$$= 4 + \sqrt{5}, \quad 4 - \sqrt{5}$$

$\uparrow$
largest eigenvalue

$\nwarrow$
smallest eigenvalue.

So, under the condition $x_1^2 + x_2^2 = 1$,

$$\left. \begin{array}{l} \text{Max. value of } 3x_1^2 + 4x_1x_2 + 5x_2^2 = 4 + \sqrt{5} \\ \text{Min. value of } 3x_1^2 + 4x_1x_2 + 5x_2^2 = 4 - \sqrt{5} \end{array} \right\} \underline{\text{Ans.}}$$

General Result: For symmetric matrix A ,

$$i) \quad \max_{\underline{x} \neq 0} \frac{\underline{x}^T A \underline{x}}{\underline{x}^T \underline{x}} = \text{Largest eigenvalue of } A.$$

$$ii) \quad \min_{\underline{x} \neq 0} \frac{\underline{x}^T A \underline{x}}{\underline{x}^T \underline{x}} = \text{Smallest eigenvalue of } A.$$

Appl.

Let, $x_1, x_2 \in \mathbb{R}$ with $x_1^2 + x_2^2 = 5$.

Find the minimum and the maximum value of $2x_1^2 + 6x_1x_2 + 3x_2^2$

Ans:

Let

$$\underline{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad \text{then } \|\underline{x}\|^2 = x_1^2 + x_2^2 = 5 \text{ (given)}$$

$$\Rightarrow \|\underline{x}\| = \sqrt{5}.$$

$$\Rightarrow \underline{x}^T \underline{x} = 5.$$

$$\text{So, given } Q(\underline{x}) = 2x_1^2 + 6x_1x_2 + 3x_2^2,$$

$$\text{The corresponding symmetric matrix} = A = \begin{bmatrix} 2 & 3 \\ 3 & 3 \end{bmatrix}$$

$$\text{So, } Q(\underline{x}) = \underline{x}^T A \underline{x}.$$

$$\text{eigenvalues of } A : \det(A - \lambda I) = 0$$

$$\Rightarrow \det \left(\begin{bmatrix} 2 & 3 \\ 3 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right) = 0$$

$$\Rightarrow \det \left(\begin{bmatrix} 2-\lambda & 3 \\ 3 & 3-\lambda \end{bmatrix} \right) = 0$$

$$\Rightarrow (2-\lambda)(3-\lambda) - 9 = 0$$

$$\Rightarrow \lambda^2 - 5\lambda + 6 - 9 = 0$$

$$\Rightarrow \lambda^2 - 5\lambda - 3 = 0$$

$$\Rightarrow \lambda = \frac{5 \pm \sqrt{25 + 12}}{2}$$

$$\Rightarrow \lambda = \frac{5 + \sqrt{37}}{2}, \quad \frac{5 - \sqrt{37}}{2}$$

$\uparrow$ Largest eigenvalue $\uparrow$ Smallest eigenvalue.

$$\text{We have, } \max_{\underline{x} \neq 0} \frac{\underline{x}^T A \underline{x}}{\underline{x}^T \underline{x}} = \text{Largest eigenvalue of } A.$$

$$\Rightarrow \max \left(\frac{2x_1^2 + 6x_1x_2 + 3x_2^2}{5} \right) = \frac{5 + \sqrt{37}}{2}$$

$$(\because x^T x = 5)$$

$$\Rightarrow \max (2x_1^2 + 6x_1x_2 + 3x_2^2) = \frac{25 + 5\sqrt{37}}{2} \quad (\text{Ans})$$

And, Similarly,

$$\min \left(\frac{2x_1^2 + 6x_1x_2 + 3x_2^2}{5} \right) = \frac{5 - \sqrt{37}}{2}$$

$$\Rightarrow \min (2x_1^2 + 6x_1x_2 + 3x_2^2) = \frac{25 - 5\sqrt{37}}{2} \quad (\text{Ans})$$

_____ / _____